

Abstract Algebra

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Introduction

I've understood abstract algebra as 3 branches: **group**, **ring**, and **field**. And therefore, I will organize topics in

Groups

Group: any set that satisfies

1. associative property

$$(ab)c = a(bc)$$

2. has identity element

$$ae = ea = a$$

3. inverse element

$$ab = ba = e$$

Example: set

$$\{1, -1, i, -i\}$$

is a group because -1 is inverse element of itself, while $i, -i$ are inverse of $-i, i$

theorem: group G 's identity is unique

proof:

$$1. \forall a \in G, ae = a$$

$$2. \forall a \in G, e'a = a$$

then

if $a = e'$, $e'e = e'$, and if $a = e$, $e'e = e$, therefore $e = e'$

theorem: since in groups, there exists an inverse element, cancellation is valid

$$ba = ca \rightarrow b = c$$

leads to **inverse element being unique** because of cancellation

given

$$ab = ba = e$$

if b and c are inverse of a , $ab = e$, and $ac = e$. therefore $b = c$

Definition: Order of a group is number of element of G expressed as $|G|$

Definition: Order of an Element is order of g , expressed as $|g|$ where g is element of G

example: group $U(15)$ where it's in mod 15,

$$U(15) = \{1, 2, 3, 4, 7, 8, 11, 13, 14\}$$

its order of group is 8 because

$$\{3\}$$

is group of units modulo 15

Definition: Subgroup is when $\forall G, \exists H$ s.t. $H \subset G$ that satisfies to be another group.

test for subgroup:

$$\forall H \leq G, \exists a, b \text{ such that } ab^{-1} \in H \text{ then } \mathbf{H \text{ is subgroup of } G}$$

proof:

since H's binary operation is same as G's, it works for associative property, for identity, $\exists x \in H$ such that $e =$

example:

$$H = \{x \in G \mid x^2 = e\}$$

let's say a, b are in H , then

$$a^2 = e, b^2 = e$$

then

$$(ab^{-1})^2 = ab^{-1}ab^{-1} = a^2(b^{-1})^2 = a^2(b^2)^{-1} = ee^{-1} = e$$

therefore by **one-step subgroup test**, **H is G's Subgroup**

Claim: $\langle a \rangle$ is a subgroup.

Proof: Since $a \in \langle a \rangle$, $\langle a \rangle \neq \emptyset$. Let $a^m, a^n \in \langle a \rangle$. Then $a^m(a^n)^{-1} = a^m a^{-n} = a^{m-n} \in \langle a \rangle$. Therefore, by the subgroup test, $\langle a \rangle \leq G$.

Cyclic Groups:

If group G can be expressed into $G = \{a^n \mid n \in \mathbb{Z}\}$, and a exists, a is generator of G , as well as cyclic group.

Test for $a^i = a^j$

Let G be a group and $a \in G$.

If a has infinite order, then

$$i = j \iff a^i = a^j.$$

If a has finite order n , then

$$\langle a \rangle = \{e, a, a^2, \dots, a^{n-1}\},$$

and

$$n \mid (i - j) \iff a^i = a^j.$$

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Theorem 1 (Fundamental Theorem of Cyclic Groups) Let $G = \langle a \rangle$ be a cyclic group with $|G| = n$. Then:

1. Every subgroup of G is cyclic.
2. The order of any subgroup of G divides n .
3. For each positive divisor k of n , there exists exactly one subgroup of order k , namely $\langle a^{n/k} \rangle$.

Every subgroup is cyclic: Let $H \leq G$. If $H = \{e\}$, it is cyclic. Otherwise, let m be the smallest positive integer such that $a^m \in H$. We claim $H = \langle a^m \rangle$.

For any $b \in H$, write $b = a^k$. By the division algorithm, $k = qm + r$ with $0 \leq r < m$. Then:

$$a^r = a^{k-qm} = a^k \cdot (a^m)^{-q} \in H$$

Since m is minimal and $0 \leq r < m$, we must have $r = 0$, so $b = (a^m)^q \in \langle a^m \rangle$. Hence $H = \langle a^m \rangle$.

Uniqueness of subgroup of order k : Let $k \mid n$. Then:

$$|\langle a^{n/k} \rangle| = \frac{n}{\gcd(n, n/k)} = k$$

If $H = \langle a^m \rangle$ has order k , then:

$$|H| = \frac{n}{\gcd(n, m)} = \frac{n}{m} = k \implies m = \frac{n}{k}$$

Thus $H = \langle a^{n/k} \rangle$ is the unique subgroup of order k .

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Theorem 2 (Number of Elements of Each Order in a Cyclic Group)
For each positive divisor d of n , the number of elements of order d in a cyclic group of order n is $\phi(d)$.

By Theorem 3.3, the group has exactly one subgroup of order d , call it $\langle a \rangle$. Every element of order d must generate $\langle a \rangle$, and by Theorem 3.2 (Corollary),

a^k generates $\langle a \rangle$ if and only if $\gcd(k, d) = 1$. Therefore the number of elements of order d is $\phi(d)$.

Remark. The number of elements of order d in a cyclic group of order n depends only on d . Thus each of Z_8 , Z_{640} , and Z_{80000} contains exactly $\phi(8) = 4$ elements of order 8.

Table of $\phi(n)$:

n	1	2	3	4	5	6	7	8	9	10	11	12
$\phi(n)$	1	1	2	2	4	2	6	4	6	4	10	4

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Definition (Group Isomorphism). An isomorphism ϕ from a group G to a group $\bar{G}$ is a bijective function from G to $\bar{G}$ that preserves the group operation. That is, for all elements $a, b \in G$:

$$\phi(ab) = \phi(a)\phi(b)$$

If an isomorphism from G to $\bar{G}$ exists, G and $\bar{G}$ are said to be isomorphic, denoted $G \approx \bar{G}$.

That isomorphic groups have the same order follows immediately from the definition. Also note that in the defining equation, the operation on the left side is that of G , while the operation on the right side is that of $\bar{G}$.

Example The map $\phi(x) = x^3$ from R to R under multiplication is *not* an isomorphism. Although ϕ is a bijection, it does not preserve the operation, since for all x, y , $(x + y)^3 \neq x^3 + y^3$ in general.

Example $U(10) \not\cong U(12)$. The proof is slightly involved. Suppose ϕ is an isomorphism from $U(10)$ to $U(12)$. Every element x of $U(12)$ satisfies $x^2 = 1$, so

$$\phi(9) = \phi(3 \cdot 3) = \phi(3)\phi(3) = 1$$

and

$$\phi(1) = \phi(1 \cdot 1) = \phi(1)\phi(1) = 1.$$

Therefore $\phi(9) = \phi(1)$. However, $9 \neq 1$, which contradicts the assumption that ϕ is injective (one-to-one).

[Coset of H in G] Let H be an arbitrary subset of a group G and let $a \in G$. Define the following sets:

$$\begin{aligned} aH &= \{ah \mid h \in H\} \\ Ha &= \{ha \mid h \in H\} \\ aHa^{-1} &= \{aha^{-1} \mid h \in H\} \end{aligned}$$

In particular, if H is a subgroup of G , then aH is called the **left coset** of H in G , and Ha is called the **right coset** of H in G . The element a is called the

example: let $G = S_3$, $H = \{(1), (13)\}$ then

$$\begin{aligned} (1)H &= H \\ (12)H &= \{(12), (12)(13)\} = \{(12), (132)\} = (132)H \\ (13)H &= \{(13), (1)\} = H \\ (23)H &= \{(23), (23)(13)\} = \{(23), (123)\} = (123)H \end{aligned}$$

example: let Z_9 's addition group $H = \{0, 3, 6\}$, then

$$\begin{aligned}
0 + H &= \{0, 3, 6\} = 3 + H = 6 + H \\
1 + H &= \{1, 4, 7\} = 4 + H = 7 + H \\
2 + H &= \{2, 5, 8\} = 5 + H = 8 + H
\end{aligned}$$

Lagrange's Theorem

For all finite group G and subgroup H , $|H|$ is a divisor of $|G|$. Furthermore, $|G|/|H|$ is the number of cosets.

proof

Set H 's different left coset as $a_1H, a_2H, \dots, a_rH$ then there exists $aH = a_iH$ for any i , since $a \in aH$, $a \in a_iH$ therefore $G = a_1H \cup a_2H \cup \dots \cup a_rH$ and since they are all different, not common multiple, $|G| = |a_1H| + |a_2H| + \dots + |a_rH|$ therefore $|G| = r|H|$

Theorem 3 *The order of an element in a finite direct product of groups is the least common multiple of the orders of its components. That is, for $(g_1, g_2, \dots, g_n) \in G_1 \oplus G_2 \oplus \dots \oplus G_n$,*

$$|(g_1, g_2, \dots, g_n)| = \text{lcm}(|g_1|, |g_2|, \dots, |g_n|)$$

Let $s = |(g_1, g_2, \dots, g_n)|$ and $t = \text{lcm}(|g_1|, |g_2|, \dots, |g_n|)$.

Since $(g_1, g_2, \dots, g_n)^t = (g_1^t, g_2^t, \dots, g_n^t) = (e_1, e_2, \dots, e_n)$, we have $s \leq t$.

Conversely, $(g_1, g_2, \dots, g_n)^s = (e_1, e_2, \dots, e_n)$ implies $g_i^s = e_i$ for all i , so $|g_i|$ divides s for each i . Hence $t \mid s$, giving $t \leq s$.

Therefore $s = t$.

Theorem 4 *For finite cyclic groups G and H , $G \oplus H$ is cyclic if and only if*

$$\gcd(|G|, |H|) = 1.$$

$$\text{Let } |G| = m, |H| = n.$$

($\Rightarrow$) Suppose $G \oplus H$ is cyclic with generator (g, h) and $\gcd(m, n) = d$. Then:

$$(g, h)^{mn/d} = \left((g^m)^{n/d}, (h^n)^{m/d} \right) = (e, e)$$

so $|(g, h)| \leq mn/d$. But $|(g, h)| = mn$ since (g, h) generates $G \oplus H$, forcing $d = 1$.

($\Leftarrow$) Suppose $\gcd(m, n) = 1$. Let $G = \langle g \rangle$, $H = \langle h \rangle$. Then:

$$|(g, h)| = \text{lcm}(m, n) = mn = |G \oplus H|$$

so (g, h) generates $G \oplus H$, making it cyclic.

Definition: Normal Subgroup is when $aH = Ha$ applies in $H \in G$

test for Normal Subgroup:

$$\forall x \in G, xHx^{-1} \subseteq H$$

proof:

this one's easy to prove, if H is normal subgroup of G , then for $x \in G$ and $h \in H$, $h' \in H$ exists such that $xh = h'x$ applies. therefore $xhx^{-1} = h'$, $xHx^{-1} \subseteq H$.

example: every subgroup of abelian group is normal subgroup($ah = ha$ applies as definition)

Definition: Group Homomorphism

$$\forall a, b \in G, \text{function } \phi \text{ where it goes } G \rightarrow \overline{G}, \phi(ab) = \phi(a) * \phi(b)$$

And Kernel

$$\{x \in G \mid \phi(x) = e\}$$

Theorem 5 *Let ϕ be a homomorphism from group G to group $\bar{G}$, and let H be a subgroup of G . Then the following hold:*

1. $\phi(H) = \{\phi(h) \mid h \in H\}$ is a subgroup of $\bar{G}$.
2. If H is cyclic, then $\phi(H)$ is also cyclic.
3. If H is abelian, then $\phi(H)$ is also abelian.
4. If H is a normal subgroup of G , then $\phi(H)$ is normal in $\phi(G)$.
5. If $|\text{Ker } \phi| = n$, then ϕ is an n -to-1 map from G onto $\phi(G)$.
6. If $|H| = n$, then $|\phi(H)|$ divides n .
7. If $\bar{K}$ is a subgroup of $\bar{G}$, then $\phi^{-1}(\bar{K}) = \{k \in G \mid \phi(k) \in \bar{K}\}$ is a subgroup of G .
8. If $\bar{K}$ is a normal subgroup of $\bar{G}$, then $\phi^{-1}(\bar{K}) = \{k \in G \mid \phi(k) \in \bar{K}\}$ is a normal subgroup of G .
9. If ϕ is surjective and $\text{Ker } \phi = \{e\}$, then ϕ is an isomorphism from G to $\bar{G}$.

Since a homomorphism preserves the group operation, properties 4, 3, 2 follow from Theorem 5.3, and properties 1, 2, 3 above correspond to properties 3, 2 of Theorem 5.3, respectively.

To prove property 4: for $\phi(h) \in \phi(H)$ and $\phi(g) \in \phi(G)$, since H is normal in G ,

$$\phi(g)\phi(h)\phi(g)^{-1} = \phi(ghg^{-1}) \in \phi(H).$$

Property 5 follows from property 6 of Theorem 9.1 and the fact that each coset of $\text{Ker } \phi = \phi^{-1}(e)$ has the same number of elements. To prove property 6: let ϕ_H denote the restriction of ϕ to H (i.e., ϕ with domain restricted to H and

codomain $\phi(H)$). By property 5, ϕ_H is a t -to-1 map, so $|\phi(H)| \cdot t = n$, which proves the property.

Since $e \in \phi^{-1}(\bar{K})$, $\phi^{-1}(\bar{K})$ is nonempty. For any $k_1, k_2 \in \phi^{-1}(\bar{K})$, we have $\phi(k_1), \phi(k_2) \in \bar{K}$, so $\phi(k_2)^{-1} \in \bar{K}$ and $\phi(k_1 k_2^{-1}) = \phi(k_1) \phi(k_2)^{-1} \in \bar{K}$. Therefore $k_1 k_2^{-1} \in \phi^{-1}(\bar{K})$, which by the one-step subgroup test proves property 7.

Property 8 is proved using the normality criterion from Theorem 9.1. Since $\bar{K}$ is normal in $\bar{G}$, for any $x \in G$ and $k \in \phi^{-1}(\bar{K})$, we have $\phi(k) \in \bar{K}$, so

$$\phi(x k x^{-1}) = \phi(x) \phi(k) (\phi(x))^{-1} \in \bar{K}.$$

Therefore $x k x^{-1} \in \phi^{-1}(\bar{K})$, proving property 8.

Finally, property 9 follows from property 5.

example: for $\phi(x) = x^4$, since $(xy)^4 = x^4 y^4$, ϕ is homomorphism, and $\text{Ker} \phi = \{x | x^4 = 1\} = \{1, -1, i, -i\}$

Theorem 6 Let $\phi : G \rightarrow \bar{G}$ be a homomorphism from group G to group $\bar{G}$. Then the map

$$\Psi : G/\text{Ker} \phi \longrightarrow \phi(G), \quad g \text{Ker} \phi \mapsto \phi(g)$$

is an isomorphism from $G/\text{Ker} \phi$ to $\phi(G)$. That is,

$$G/\text{Ker} \phi \cong \phi(G).$$

Call the map $g \text{Ker} \phi \mapsto \phi(g)$ by the name Ψ . By property 5 of Theorem 10.1, Ψ is well-defined (i.e. its value is independent of the choice of coset representative) and is injective. Moreover,

$$\begin{aligned} \Psi(x \text{Ker} \phi \cdot y \text{Ker} \phi) &= \Psi(xy \text{Ker} \phi) \\ &= \phi(xy) \\ &= \phi(x) \phi(y) \\ &= \Psi(x \text{Ker} \phi) \Psi(y \text{Ker} \phi), \end{aligned}$$

so Ψ preserves the group operation and is therefore a homomorphism. Since Ψ is both injective and surjective onto $\phi(G)$, it is an isomorphism.

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Theorem 7 (Existence of Subgroups of Prime-Power Order (Sylow's First Theorem, 1872))

Let G be a finite group and p a prime. If p^k divides $|G|$, then G has at least one subgroup of order p^k .

We apply mathematical induction on $|G|$. If $|G| = 1$, Theorem 11.3 is trivially true. Now assume the theorem holds for all groups of order less than $|G|$.

If G has a proper subgroup H such that p^k divides $|H|$, then by the induction hypothesis H contains a subgroup of order p^k , and the theorem is proved.

Therefore, assume that no proper subgroup of G has order divisible by p^k .

Consider the class equation of G :

$$|G| = |Z(G)| + \sum |G : C(a)|,$$

where the sum is taken over one representative a from each conjugacy class $\text{cl}(a)$ with $a \notin Z(G)$.

Since p^k divides $|G| = |G : C(a)| \cdot |C(a)|$ and p^k does not divide $|C(a)|$, we have that p divides $|G : C(a)|$ for every $a \notin Z(G)$. From the class equation it follows that p divides $|Z(G)|$.

By the Fundamental Theorem of Finite Abelian Groups (Theorem 10.1), or by Theorem 9.5, $Z(G)$ has an element x of order p . Since x is a central element of G , $\langle x \rangle$ is a normal subgroup of G and we can form the quotient group $G/\langle x \rangle$.

Now p^{k-1} divides $|G/\langle x \rangle|$, so by the induction hypothesis $G/\langle x \rangle$ has a subgroup of order p^{k-1} . By Exercise 21 of Chapter 9, this subgroup has the form $H/\langle x \rangle$ for some subgroup H of G .

Finally, since $|H/\langle x \rangle| = p^{k-1}$ and $|\langle x \rangle| = p$, we get $|H| = p^k$. This completes the proof.

Theorem 8 (Sylow's Second Theorem) *If H is a subgroup of a finite group G and $|H|$ is a power of a prime p , then H is contained in some Sylow p -subgroup of G .*

Let K be a Sylow p -subgroup of G , and let $\mathcal{C} = \{K_1, K_2, \dots, K_n\}$ be the set of all conjugates of K in G , where $K = K_1$.

Let $S_{\mathcal{C}}$ be the set of all permutations of $\mathcal{C}$. For each $g \in G$, define $\phi_g : \mathcal{C} \rightarrow \mathcal{C}$ by

$$\phi_g(K_i) = gK_i g^{-1}.$$

One easily verifies that $\phi_g \in S_{\mathcal{C}}$.

Define a map $T : G \rightarrow S_{\mathcal{C}}$ by $T(g) = \phi_g$. Since

$$\phi_{gh}(K_i) = (gh)K_i(gh)^{-1} = g(hK_i h^{-1})g^{-1} = \phi_g(\phi_h(K_i)) = (\phi_g \circ \phi_h)(K_i),$$

we have $\phi_{gh} = \phi_g \circ \phi_h$, so T is a homomorphism from G to $S_{\mathcal{C}}$.

Consider the image $T(H)$ of H under T . Since $|H|$ is a power of p , so is $|T(H)|$ (see property 6 of Theorem 9.2). By the Orbit-Stabilizer Theorem (Theorem 6.3), $|\text{orb}_{T(H)}(K_i)|$ divides $|T(H)|$ for each i , so $|\text{orb}_{T(H)}(K_i)|$ is a power of p .

Now consider when $|\text{orb}_{T(H)}(K_i)| = 1$. This means $\phi_g(K_i) = gK_i g^{-1} = K_i$ for all $g \in H$, i.e., $H \leq N(K_i)$.

The elements of $N(K_i)$ whose order is a power of p are exactly the elements of K_i (see Exercise 5). Therefore, the necessary and sufficient condition for $|\text{orb}_{T(H)}(K_i)| = 1$ is $H \leq K_i$.

To complete the proof, we need to show that $|\text{orb}_{T(H)}(K_i)| = 1$ for some i . By Theorem 11.1, $|\mathcal{C}| = |G : N(K)|$ (see Exercise 11). Since $|G : K| = |G : N(K)| \cdot |N(K) : K|$ is not divisible by p , $|\mathcal{C}|$ is also not divisible by p .

Since the orbits partition $\mathcal{C}$, $|\mathcal{C}|$ is the sum of orbit sizes, all of which are powers of p . If no orbit had size 1, then p would divide $|\mathcal{C}|$, a contradiction. Therefore an orbit of size 1 exists, and the proof is complete.

Theorem 9 (Sylow's Third Theorem) *Let p be a prime and G a group of order $p^n m$, where p does not divide m . Then the number n_p of Sylow p -subgroups of G satisfies:*

1. $n_p \equiv 1 \pmod{p}$,
2. n_p divides m ,
3. any two Sylow p -subgroups of G are conjugate.

Let K be a Sylow p -subgroup of G and let $\mathcal{C} = \{K_1, K_2, \dots, K_n\}$ be the set of all conjugates of K in G , where $K = K_1$.

Step 1: $n \equiv 1 \pmod{p}$. Define $S_{\mathcal{C}}$, T , and the action of K on $\mathcal{C}$ via T exactly as in the proof of Theorem 11.4, but now consider the image $T(K)$ instead of $T(H)$.

As before, $|\text{orb}_{T(K)}(K_i)|$ is a power of p , and $|\text{orb}_{T(K)}(K_i)| = 1$ if and only if $K \leq K_i$. Since K and K_i are both Sylow p -subgroups (hence of equal order), $K \leq K_i$ implies $K = K_i$. Therefore the unique orbit of size 1 is $\{K_1\} = \{K\}$, and every other orbit has size a multiple of p greater than 1.

Since the orbits partition $\mathcal{C}$, we get $n = |\mathcal{C}| \equiv 1 \pmod{p}$.

Step 2: Every Sylow p -subgroup belongs to $\mathcal{C}$. Suppose H is a Sylow p -subgroup not in $\mathcal{C}$. Using $T(H)$ as before, $|\mathcal{C}|$ is the sum of orbit sizes under $T(H)$, each a power of p . Since $H \notin \mathcal{C}$, no orbit has size 1, so p divides $|\mathcal{C}|$. But from Step 1, $|\mathcal{C}| \equiv 1 \pmod{p}$, a contradiction. Hence $H \in \mathcal{C}$, and $\mathcal{C}$ is exactly the set of all Sylow p -subgroups, so $n_p = |\mathcal{C}|$.

Step 3: n_p divides m . This follows directly from $n_p = |\mathcal{C}| = |G : N(K)|$ (see Exercise 11). $\square$

Ring

Ring is when this applies:

1. $a + b = b + a$
2. $(a + b) + c = a + (b + c)$
3. identity in respect to addition
4. $\forall a \in R, \exists -a \in R, a + (-a) = 0$
5. $a(bc) = (ab)c$
6. $a(b + c) = ab + ac, (b + c)a = ba + ca$

Example:

$\mathbb{Z}$ is both addition and multiplication wise identity existing, and $\{1, -1\}$

Test for Subring if

$$a - b, ab \in S \forall a, b$$

they are subring.

Ideal and factor rings:

it is called ideal when

$$\forall r \in R, a \in A \exists ra \in A, ar \in A$$

Test for Ideal

1. $\forall a, b \in A, a - b \in A$
2. $\forall a \in A$ and $r \in R, ra \in A$ and $ar \in A$

Example: for

$$nZ = \{0, \pm n, \pm 2n, \dots\}$$

is Z 's Ideal

Factor Ring's existence

if R is ring and A is R 's subring, then $\{r + a | r \in R\}$ being $(s + A) + (t + A) = (s + t) + A$, $(s + A)(t + A) = st + A$ is equivalent for A being R 's Ideal

example:

$$Z/4Z = \{0 + 4Z, 1 + 4Z, 2 + 4Z, 3 + 4Z\}$$

then for two elements,

$$\begin{aligned}(2 + 4Z) + (3 + 4Z) &= 5 + 4Z = 1 + 4Z \\(2 + 4Z)(3 + 4Z) &= 6 + 4Z = 2 + 4Z\end{aligned}$$

Definition

it is called prime ideal when

$$\forall a, b \in R, \text{ if } ab \in A, \text{ it's } a \in A \text{ or } b \in A$$

it is called maximal ideal when

$$\text{if } A \subseteq B \subseteq R \text{ then } B = A \text{ or } B = R$$

Definition: It is called ring homomorphism when

$$\phi(a + b) = \phi(a) + \phi(b), \phi(ab) = \phi(a)\phi(b)$$

When it's injective and bijective, it is called ring isomorphism.

example:

ϕ is ring homomorphism on behalf of $Z_4 \rightarrow Z_{10}$ on $x \rightarrow 5x$ because

$$\phi(x + y) = \phi(x) + \phi(y), 5(x + y) = 5x + 5y$$

and

$$5xy = 5x * 5y$$

on Z_{10}

Theorem 10 (Properties of Ring Homomorphisms) *Let ϕ be a ring homomorphism from a ring R to a ring S . Let A be a subring of R and B be an ideal of S .*

1. *For any $r \in R$ and any positive integer n ,*

$$\phi(nr) = n\phi(r) \quad \text{and} \quad \phi(r^n) = (\phi(r))^n.$$

2. *$\phi(A) = \{\phi(a) \mid a \in A\}$ is a subring of S .*
3. *If A is an ideal and ϕ is onto S , then $\phi(A)$ is an ideal.*
4. *$\phi^{-1}(B) = \{r \in R \mid \phi(r) \in B\}$ is an ideal of R .*
5. *If R is commutative, then $\phi(R)$ is commutative.*
6. *If R has unity 1 and $S \neq \{0\}$ and ϕ is onto, then $\phi(1)$ is the unity of S .*
7. *ϕ is an isomorphism if and only if ϕ is onto and*

$$\ker \phi = \{r \in R \mid \phi(r) = 0\} = \{0\}.$$

8. *If ϕ is an isomorphism from R to S , then ϕ^{-1} is an isomorphism from S to R .*

The proofs follow by the same methods as in Theorems 9.1 and 9.2.

Polynomial Rings

if R is abelian Ring, then

$$R[x] = \{a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 \mid a_i \in R\}$$

is called polynomials in terms of x

Theorem 11 (D an Integral Domain Implies $D[x]$ is an Integral Domain) *If D is an integral domain, then*

We already know $D[x]$ is a commutative ring with unity and no zero-divisors. Let

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad g(x) = b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0,$$

with $a_n \neq 0$, $b_m \neq 0$. The leading term of $f(x)g(x)$ is $a_n b_m$, and since D is an integral domain, $a_n b_m \neq 0$.

Theorem 12 (Division Algorithm for $F[x]$) *Let F be a field and $f(x), g(x) \in F[x]$ with $g(x) \neq 0$. Then there exist unique $q(x), r(x) \in F[x]$ such that*

$$f(x) = g(x)q(x) + r(x),$$

where $r(x) = 0$ or $\deg r(x) < \deg g(x)$.

By the Well-Ordering Principle applied to $\deg f(x) - \deg g(x)$, we induct on $\deg f(x)$. If $f(x) = 0$ or $\deg f(x) < \deg g(x)$, take $q(x) = 0$ and $r(x) = f(x)$.

Otherwise, let $\deg f(x) = n$, $\deg g(x) = m$, and write

$$f(x) = a_n x^n + \cdots + a_0, \quad g(x) = b_m x^m + \cdots + b_0.$$

Set $f_1(x) = f(x) - a_n b_m^{-1} x^{n-m} g(x)$, so $\deg f_1(x) < \deg f(x)$. By induction, $f_1(x) = g(x)q_1(x) + r(x)$, giving

$$f(x) = g(x)(a_n b_m^{-1} x^{n-m} + q_1(x)) + r(x).$$

Uniqueness follows from the fact that if two such representations existed, their difference would yield a contradiction on degrees.

Divide $f(x) = 3x^4 + x^3 + 2x^2 + 1$ by $g(x) = x^2 + 4x + 2$ in $Z_5[x]$:

$$x^2 + 4x + 2 \mid 3x^4 + x^3 + 2x^2 + 1 \longrightarrow q(x) = 3x^2 + 4x, \quad r(x) = 2x + 1.$$

Thus $3x^4 + x^3 + 2x^2 + 1 = (x^2 + 4x + 2)(3x^2 + 4x) + (2x + 1)$.

Definition: it is called Principal Ideal Domain, PID when the Ideal is in form of

$$\langle a \rangle = \{ra \mid r \in R\}$$

Theorem if F is a field, $F[x]$ is a PID

proof as the theorem states, $F[x]$ is Ideal Domain.

say I is Ideal of $F[x]$, then $g(x) \in I$, $\langle g(x) \rangle \subseteq I$, then $\deg r(x) < \deg g(x)$ so $f(x) \in \langle g(x) \rangle$, $I \subseteq \langle g(x) \rangle$

[Irreducible / Reducible Polynomial] Let D be an integral domain. A non-zero, non-unit $f(x) \in D[x]$ is **irreducible** over D if whenever $f(x) = g(x)h(x)$ with $g(x), h(x) \in D[x]$, then $g(x)$ or $h(x)$ is a unit in $D[x]$. Otherwise, $f(x)$ is **reducible**.

In particular, for a field F , a non-constant $f(x) \in F[x]$ is irreducible iff $f(x)$ cannot be expressed as a product of two polynomials of lower degree.

example: $f(x) = 2x^2 + 4$ is irreducible over Q , but reducible over Z , since

$$2x^2 + 4 = 2(x^2 + 2),$$

where neither 2 nor $x^2 + 2$ is a unit in $Z[x]$.

theorem

Let F be a field and $f(x) \in F[x]$ being $\deg f(x) = 2, 3$ then $f(x)$ being reducible in F is equivalent as $f(x)$ having root in F

proof:

if $f(x) = g(x)h(x)$, then $\deg f(x) = \deg g(x) + \deg h(x)$, and since they are 2 or 3, either $g(x)$ or $h(x)$ will be 1-degree polynomial. This also means if $g(x) = ax + b$, then $g(x)$'s root is $-a^{-1}b$, also being $f(x)$'s root. inversely, if $f(a) = 0$ as $a \in F$, as factorization theorem, $x - a$ will be $f(x)$'s factor. therefore $f(x)$ is F 's reducible polynomial

Theorem 13 (Eisenstein's Criterion) Let $f(x) = a_n x^n + \dots + a_0 \in Z[x]$. If a prime p satisfies

$$p \mid a_n, \quad p \mid a_{n-1}, \dots, p \mid a_0, \quad p^2 \nmid a_0,$$

then $f(x)$ is irreducible over Q .

Suppose $f(x) = g(x)h(x)$ with $g(x) = b_r x^r + \cdots + b_0$ and $h(x) = c_s x^s + \cdots + c_0$. Since $p \mid a_0 = b_0 c_0$ but $p^2 \nmid a_0$, say $p \mid b_0$ and $p \nmid c_0$. Since $p a_n = b_r c_s$, we have $p \nmid b_r$, so let t be the smallest index with $p \nmid b_t$. From $a_t = b_t c_0 + b_{t-1} c_1 + \cdots + b_0 c_t$, every term but $b_t c_0$ is divisible by p , so $p \mid a_t$. But $t < n$, contradicting $p \nmid a_t$.

[p -th Cyclotomic Polynomial] For any prime p ,

$$\Phi_p(x) = x^{p-1} + x^{p-2} + \cdots + 1$$

is irreducible over Q .

Set $f(x) = \Phi_p(x+1) = x^{p-1} + \binom{p}{1}x^{p-2} + \cdots + \binom{p}{p-1}$. All coefficients except the leading one are divisible by p , and the constant term p satisfies $p^2 \nmid p$. By Eisenstein, $f(x)$ is irreducible, hence so is $\Phi_p(x)$.

Theorem 14 $\langle p(x) \rangle$ is maximal in $F[x]$ iff $p(x)$ is irreducible over F .

($\Rightarrow$) Suppose $\langle p(x) \rangle$ is maximal. If $p(x) = g(x)h(x)$, then $\langle p(x) \rangle \subseteq \langle g(x) \rangle \subseteq F[x]$, so either $\langle g(x) \rangle = \langle p(x) \rangle$ or $\langle g(x) \rangle = F[x]$, forcing $\deg g(x) = \deg p(x)$ or $\deg g(x) = 0$. So $p(x)$ is irreducible.

($\Leftarrow$) Suppose $p(x)$ is irreducible and $\langle p(x) \rangle \subseteq I \subseteq F[x]$. Since $F[x]$ is a PID, $I = \langle g(x) \rangle$ and $p(x) = g(x)h(x)$. By irreducibility, $g(x)$ or $h(x)$ is constant, giving $I = F[x]$ or $I = \langle p(x) \rangle$. Thus $\langle p(x) \rangle$ is maximal.

If $p(x)$ is irreducible over F , then $F[x]/\langle p(x) \rangle$ is a field.

extension field: when

$$F \subseteq E$$

that E 's operation is limited on F 's, it's called as E is F 's **Extension Field**

example: given

$$f(x) = x^5 + 2x^2 + 2x + 2 \in Z_3[x]$$

then factorization is

$$(x^2 + 1)(x^3 + 2x + 2)$$

using this fact,

$$E = Z_3[x] / \langle x^2 + 1 \rangle$$

number of element is 9, or

$$E = Z_3[x] / \langle x^3 + 2x + 2 \rangle$$

has element of 27

Definition: Splitting field when $f(x)$ is factorized by $E[x]$ as 1st order, it's called splitting field

example: given

$$f(x) = x^2 + 1 \in Q[x],$$

this factorizes to

$$x^2 + 1 = (x + \sqrt{-1})(x - \sqrt{-1})$$

and therefore this splits in C

amsart amsmath, amsthm, amssymb

Theorem 15 (Fundamental Theorem of Galois Theory) *Let F be a field of characteristic 0 or finite, E/F a splitting field of some $f(x) \in F[x]$, $G = \text{Gal}(E/F)$. For K with $F \subseteq K \subseteq E$, define $\phi : \mathcal{F} \rightarrow \mathcal{G}$ by $\phi(K) = \text{Gal}(E/K)$. Then:*

1. $[E : K] = |\text{Gal}(E/K)|$ and $[K : F] = |G|/|\text{Gal}(E/K)|$.
2. K/F Galois $\Rightarrow \text{Gal}(E/K)G$ and $\text{Gal}(K/F) \cong G/\text{Gal}(E/K)$.
3. $K = E_{\text{Gal}(E/K)}$.
4. $H \leq G \Rightarrow E_H$ is fixed by exactly H , i.e., $\text{Gal}(E/E_H) = H$.

(1) By induction on $[E : F] = n$.

Base: $n = 1 \Rightarrow E = F$, trivial.

Inductive step: Let $[E : F] = n \geq 2$, E a splitting field of $f(x) \in F[x]$. Let $p(x) \mid f(x)$ be irreducible of degree $m = \deg(p) \geq 2$. Since $\text{char}(F) = 0$ or F finite, $p(x)$ has m distinct roots $u = u_1, \dots, u_m \in E$.

Let $G = \text{Gal}(E/F)$, $R = \{u_1, \dots, u_m\}$. For each u_i , $\sigma \mapsto u_i$ is an algebraic conjugate of $p(x)$, so $R = \{\sigma(u) \mid \sigma \in G\}$.

Set $H = \{\sigma \in G \mid \sigma(u) = u\}$, $X = \{\sigma H \mid \sigma \in G\}$. Then $H \leq G$, $|G| = [G : H]|H| = |X||H|$.

Note: $\sigma H = \tau H \Leftrightarrow \tau^{-1}\sigma \in H \Leftrightarrow \sigma(u) = \tau(u)$. So $\phi : X \rightarrow R$, $\phi(\sigma H) = \sigma(u)$ is a bijection, giving $|X| = |R| = m$.

Hence $|G| = m|H| = [F(u) : F]|H|$.

Since $\sigma \in \text{Gal}(E/F(u))$ fixes u , we get $\text{Gal}(E/F(u)) \subseteq H$. Conversely, $\sigma \in H \Rightarrow \sigma(u) = u$ and each $v \in F(u)$ writes as $v = a_0 + a_1u + \dots + a_{m-1}u^{m-1}$, so $\sigma(v) = v$, giving $H \subseteq \text{Gal}(E/F(u))$.

Thus $H = \text{Gal}(E/F(u))$ and $|H| = |\text{Gal}(E/F(u))| = [E : F(u)]$ (induction hypothesis, since $[E : F(u)] < n$).

Therefore $|G| = [F(u) : F][E : F(u)] = [E : F] = n$, completing (1).

Also, $[K : F] = |G|/|\text{Gal}(E/K)|$ follows since $\sigma\text{Gal}(E/K) = \tau\text{Gal}(E/K) \Leftrightarrow \tau^{-1}\sigma \in \text{Gal}(E/K) \Leftrightarrow \sigma(\alpha) = \tau(\alpha) \forall \alpha \in K$, so $[K : F] = |G|/|\text{Gal}(E/K)|$.

(3) & (4) By definition $K \subseteq E_{\text{Gal}(E/K)}$. For $\alpha \in E \setminus K$: since E/F is a splitting field, $\exists \sigma \in \text{Gal}(E/K)$ with $\sigma(\alpha) \neq \alpha$, so $\alpha \notin E_{\text{Gal}(E/K)}$. Thus $E_{\text{Gal}(E/K)} = K$.

For (4): $H \leq \text{Gal}(E/E_H)$ by definition. Suppose $|H| < |\text{Gal}(E/E_H)| = n$; set $H = \{\sigma_1, \dots, \sigma_{|H|}\}$ and $f(x) = \prod_{i=1}^{|H|} (x - \sigma_i(\alpha))$ for some $\alpha \in K$. Coefficients of f are symmetric in $\sigma_i(\alpha)$'s, hence in E_H . But $f(\alpha) = 0$ gives $\deg(\alpha, E_H) \leq |H| < n = [E : E_H]$, contradiction. So $H = \text{Gal}(E/E_H)$.

(2) K/F Galois $\Rightarrow K$ is a splitting field of some $g(x) \in F[x]$. By (3), $K = E_{\text{Gal}(E/K)}$, and for $\sigma \in \text{Gal}(E/F)$, $\alpha \in K$: $\sigma(\alpha) \in K$, so $\tau(\sigma(\alpha)) = \sigma(\alpha)$ for all $\tau \in \text{Gal}(E/K)$, i.e., $(\sigma^{-1}\tau\sigma)(\alpha) = \alpha$, giving $\text{Gal}(E/K)G$.

Define $\phi : G \rightarrow \text{Gal}(K/F)$ by $\phi(\sigma) = \sigma_K$ (restriction). $\ker(\phi) = \text{Gal}(E/K)$, so by the first isomorphism theorem,

$$\text{Gal}(K/F) \cong G/\text{Gal}(E/K).$$